

TASK 2: ANALYZE A PHISHING EMAIL SAMPLE

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Task Title: Analysis of a Phishing Email Sample

Objective

The objective of this task is to analyze a suspicious email and identify phishing characteristics such as spoofed sender information, suspicious email headers, social engineering techniques, spam indicators, and other signs commonly used in phishing attacks.

Tools Used

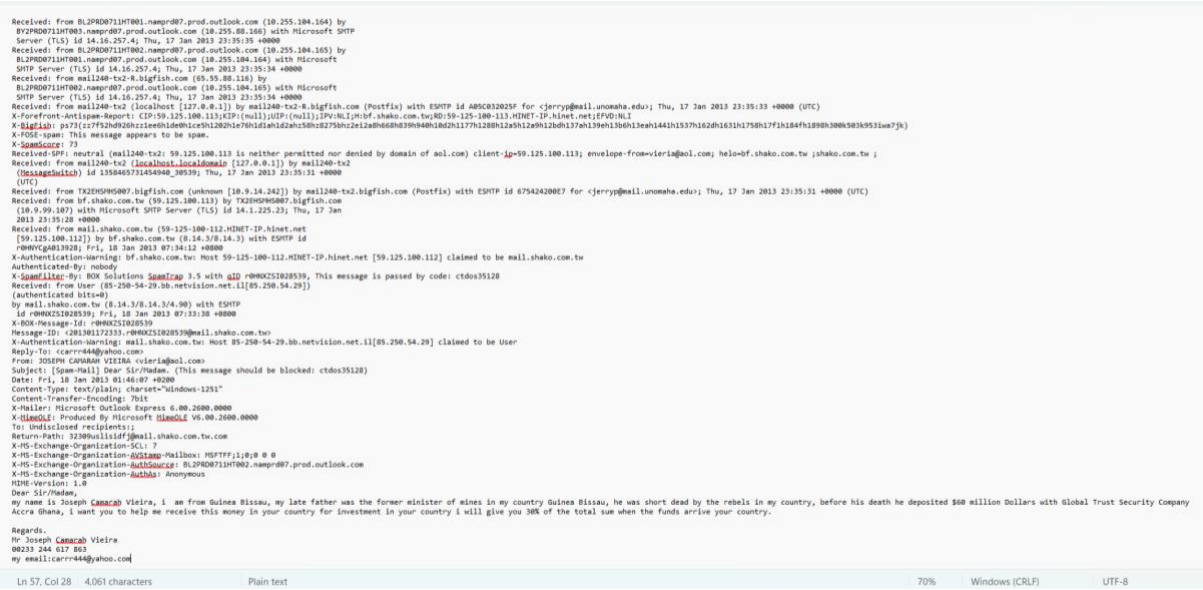
- 1. Notepad
- 2. Microsoft Word
- 3. Email Header Analysis
- 4. Sample Phishing Email (.txt file)

Procedure

Step 1: Obtain a Sample Phishing Email

A phishing email sample was obtained and saved as a text file named "phishing_email.txt" for analysis.

Screenshot 1:



Step 2: Examine Sender's Email Address

The sender information was analyzed.

Sender Address:

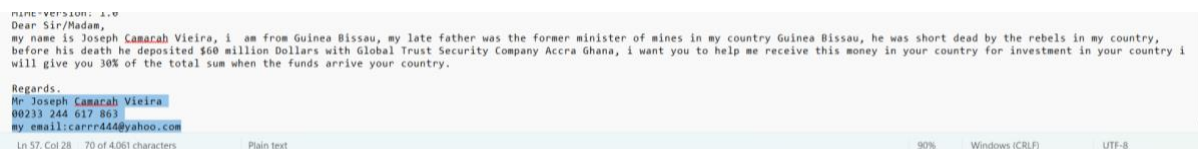
From: vieria@aol.com

Reply-To Address: carr444@yahoo.com

Observation: The sender email address and Reply-To email address belong to different domains. Legitimate organizations generally use consistent email domains for communication.

Finding: Suspicious sender identity and Reply-To mismatch detected.

Screenshot 2:



Dear Sir/Madam,
my name is Joseph Camarab Vieira, i am from Guinea Bissau, my late father was the former minister of mines in my country Guinea Bissau, he was short dead by the rebels in my country, before his death he deposited \$60 million Dollars with Global Trust Security Company Accra Ghana, i want you to help me receive this money in your country for investment in your country i will give you 30% of the total sum when the funds arrive your country.

Regards,
Mr Joseph Camarab Vieira
00233 244 617 863
my email: carr444@yahoo.com

Ln 57, Col 28 70 of 4061 characters Plain text 90% Windows (CRLF) UTF-8

Step 3: Analyze Email Headers

The email headers were examined for spam indicators and authentication information.

Important Header Information:

X-FOSE-spam: This message appears to be spam

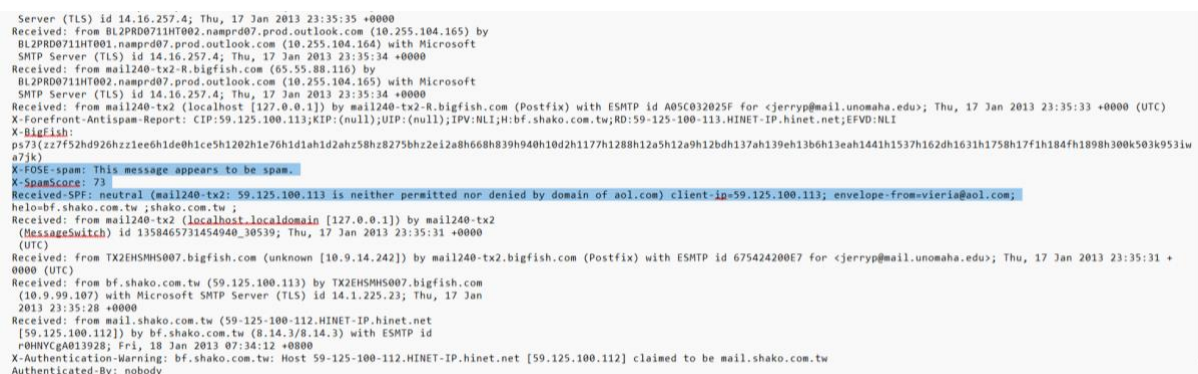
X-SpamScore: 73

Received-SPF: neutral

Observation: The email was automatically flagged as spam by the mail security system. A spam score of 73 indicates a highly suspicious message. SPF verification was neutral, meaning the sender could not be fully authenticated.

Finding: High spam score and suspicious authentication results detected.

Screenshot 3:



Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:35 +0000
Received: from BL2PRD071HT002.namprd07.prod.outlook.com (10.255.104.165) by BL2PRD071HT001.namprd07.prod.outlook.com (10.255.104.164) with Microsoft SMTP Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:34 +0000
Received: from mail240-tx2-R.bigfish.com (65.55.88.116) by BL2PRD071HT002.namprd07.prod.outlook.com (10.255.104.165) with Microsoft SMTP Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:34 +0000
Received: from mail240-tx2 [localhost [127.0.0.1]] by mail240-tx2-R.bigfish.com (Postfix) with ESMTP id A05C032025F for <jerry@mail.unomaha.edu>; Thu, 17 Jan 2013 23:35:33 +0000 (UTC)
X-forefront-Antispam-Report: CIP:59.125.100.113;KIP:(null);UIP:(null);IPV:NLI;W:bf.shako.com.tw;RD:59-125-100-113.HINET-IP.hinet.net;EFVD:NLI
X-Bigfish: ps73(zz7f52hd926hz1ee6h1de0h1ce5h1202h1e76h1d1ah1d2ah258hz8275bh2ze12a8h668h839h940h10d2h1177h1288h12a5h12a9h12bdh137ah139eh13b6h13eah1441h1537h162dh1631h1758h17f1h184fh1898h300k503k953iw47k)
X-FOSE-spam: This message appears to be spam.
X-SpamScore: 73
Received-SPF: neutral (mail240-tx2: 59.125.100.113 is neither permitted nor denied by domain of aol.com) client-ip=59.125.100.113; envelope-from=vieria@aol.com; helo=bf.shako.com.tw; shako.com.tw;
Received: from mail240-tx2 [localhost [127.0.0.1]] by mail240-tx2 (MessageSwitch) id 1358465731454940_30539; Thu, 17 Jan 2013 23:35:31 +0000 (UTC)
Received: from TX2EHS07.bigfish.com (unknown [10.9.14.242]) by mail240-tx2.bigfish.com (Postfix) with ESMTP id 675424200E7 for <jerry@mail.unomaha.edu>; Thu, 17 Jan 2013 23:35:31 +0000 (UTC)
Received: from bf.shako.com.tw (59.125.100.113) by TX2EHS07.bigfish.com (10.9.99.107) with Microsoft SMTP Server (TLS) id 14.1.225.23; Thu, 17 Jan 2013 23:35:28 +0000
Received: from mail.shako.com.tw (59-125-100-112.HINET-IP.hinet.net [59.125.100.112]) by bf.shako.com.tw (8.14.3/8.14.3) with ESMTP id r0HNVGpA013928; Fri, 18 Jan 2013 07:34:12 +0000
X-Authentication-Warning: bf.shako.com.tw: Host 59-125-100-112.HINET-IP.hinet.net [59.125.100.112] claimed to be mail.shako.com.tw
Authenticated-By: nobody

Step 4: Identify Suspicious Links or Attachments

The email content was reviewed for malicious links and file attachments.

Observation

No hyperlinks were found in the email.

No attachments were included with the email.

Finding: No suspicious attachments or URLs were detected.

Step 5: Analyze Email Content

The email claims that the sender is the son of a former government minister and requests assistance in transferring \$60 million in exchange for 30% of the funds.

Observation: The message promises a large financial reward to gain the recipient's trust. This technique is commonly used in advance-fee fraud and phishing scams.

Finding: Financial scam and social engineering techniques detected.

Screenshot 4:

```
Content-Transfer-Encoding: 7bit
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000
To: Undisclosed recipients;
Return-Path: 32309uslisdj@mail.shako.com.tw.com
X-MS-Exchange-Organization-SCL: 7
X-MS-Exchange-Organization-AVStamp-Mailbox: MSFTFF;1;0;0 0 0
X-MS-Exchange-Organization-AuthSource: BL2PRD0711H002.namprd07.prod.outlook.com
X-MS-Exchange-Organization-AuthAs: Anonymous
MIME-Version: 1.0
Dear Sir/Madam,
my name is Joseph Camarah Vieira, i am from Guinea Bissau, my late father was the former minister of mines in my country Guinea Bissau, he was short dead by the rebels in my country,
before his death he deposited $60 million Dollars with Global Trust Security Company Accra Ghana, i want you to help me receive this money in your country for investment in your country i
will give you 30% of the total sum when the funds arrive your country.
Regards.
Mr Joseph Camarah Vieira
00233 244 617 863
my email:carr444@yahoo.com
```

Step 6: Check for Mismatched URLs

The email was examined for visible URLs and domain mismatches.

Observation: No URLs were present in the email body.

Finding: No URL mismatch detected.

Step 7: Verify Spelling and Grammar Errors

The email text was reviewed for language quality.

Examples:

"he was short dead by the rebels"

"when the funds arrive your country"

Observation: Multiple grammar and spelling errors were identified. Such errors are frequently found in phishing and scam emails.

Finding: Grammar and language inconsistencies detected.

Screenshot 5:

```
Reply-To: <carr44@yahoo.com>
From: JOSEPH CAMARAH VIEIRA <vieira@aol.com>
Subject: [Spam-Mail] Dear Sir/Madam. (This message should be blocked: ctdos35128)
Date: Fri, 18 Jan 2013 01:46:07 +0200
Content-Type: text/plain; charset="Windows-1251"
Content-Transfer-Encoding: 7bit
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000
To: Undisclosed recipients;
Return-Path: 32309uslisdj@mail.shako.com.tw.com
X-MS-Exchange-Organization-SCL: 7
X-MS-Exchange-Organization-AVStamp-Mailbox: MSFTFF;1;0;0 0 0
X-MS-Exchange-Organization-AuthSource: BL2PRD0711HT002.namprd07.prod.outlook.com
X-MS-Exchange-Organization-AuthAs: Anonymous
MIME-Version: 1.0
Dear Sir/Madam,
my name is Joseph Camarah Vieira, i am from Guinea Bissau, my late father was the former minister of mines in my country Guinea Bissau, he was short dead by the rebels in my country,
before his death he deposited $60 million dollars with Global Trust Security Company Accra Ghana, i want you to help me receive this money in your country for investment in your country i
will give you 30% of the total sum when the funds arrive your country.

Regards,
Mr Joseph Camarah Vieira
00233 244 617 863
my email:carr44@yahoo.com
```

Step 8: Summary of Phishing Indicators

Indicator	Result
Sender Address Mismatch	Yes
Suspicious Reply-To Address	Yes
Spam Flagged by Mail Server	Yes
High Spam Score	Yes
SPF Authentication Issue	Yes
Financial Scam Content	Yes
Social Engineering Attempt	Yes
Grammar Errors	Yes
Suspicious URL	No
Attachment Present	No

Analysis Results

The email demonstrates several characteristics commonly associated with phishing and scam campaigns.

Major indicators identified include:

1. Mismatch between sender and Reply-To addresses.
2. High spam score generated by email security systems.
3. Neutral SPF authentication result.

4. Promise of a large financial reward.
5. Use of social engineering techniques.
6. Multiple spelling and grammar errors.
7. Email classified as spam by mail filters.

These indicators strongly suggest that the email is fraudulent and should not be trusted.

Conclusion

The email was successfully analyzed for phishing characteristics. Multiple indicators of phishing and spam activity were identified, including sender identity inconsistencies, spam detection warnings, financial scam content, and language errors. Based on the analysis, the email is classified as a phishing/scam email intended to deceive recipients into engaging with fraudulent activities.

Outcome

Successfully learned how to:

- Analyze suspicious email messages
- Examine email headers
- Identify phishing indicators
- Detect sender spoofing attempts
- Recognize social engineering tactics
- Evaluate email authenticity
- Document phishing analysis findings